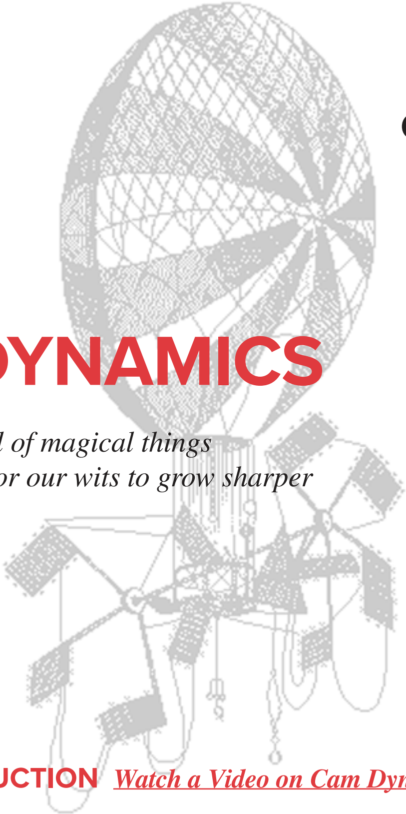
Chapter 15



CAM DYNAMICS

The universe is full of magical things
patiently waiting for our wits to grow sharper
EDEN PHILLPOTS

15.0 INTRODUCTION *Watch a Video on Cam Dynamics (48:29)*[*]

Chapter 8 presented the kinematics of cams and followers and methods for their design. We will now extend the study of cam-follower systems to include considerations of the dynamic forces and torques developed. While the discussion in this chapter is limited to examples of cams and followers, the principles and approaches presented are applicable to most dynamic systems. The cam-follower system can be considered a useful and convenient example for the presentation of topics such as creating lumped parameter dynamic models and defining equivalent systems as described in Chapter 10. These techniques as well as the discussion of natural frequencies, effects of damping, and analogies between physical systems will be found useful in the analysis of all dynamic systems regardless of type.

In Chapter 10 we discussed the two approaches to dynamic analysis, commonly called the forward and the inverse dynamics problems. The forward problem assumes that all the forces acting on the system are known and seeks to solve for the resulting displacements, velocities, and accelerations. The inverse problem is, as its name says, the inverse of the other. The displacements, velocities, and accelerations are known, and we solve for the dynamic forces that result. In this chapter we will explore the application of both methods to cam-follower dynamics. Section 15.1 explores the forward solution. Section 15.3 will present the inverse solution. Both are instructive in this application of a force-closed (spring-loaded) cam-follower system and will each be discussed in this chapter.

[*] http://www.designofmachinery.com/DOM/Cam_Dynamics.mp4

TABLE 15-1 Notation Used in This Chapter

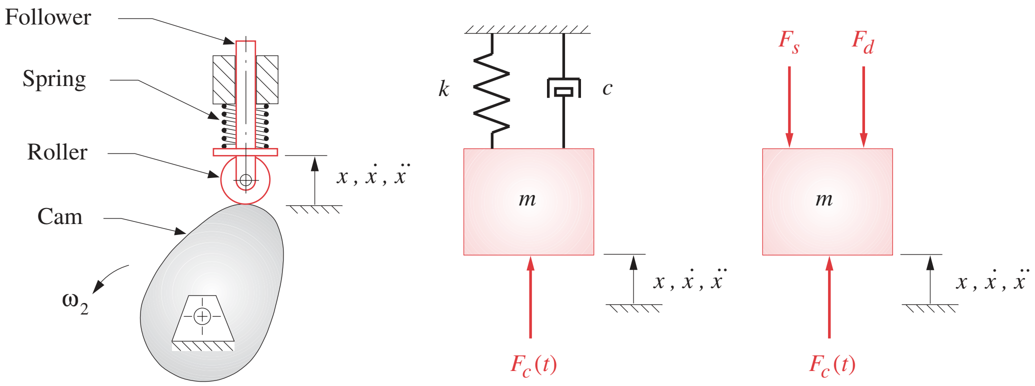
c = *damping coefficient*
c_c = *critical damping coefficient*
k = *spring constant*
F_c = *force of cam on follower*
F_s = *force of spring on follower*
F_d = *force of damper on follower*
m = *mass of moving elements*
t = *time in seconds*
T_c = *torque on camshaft*
θ = *camshaft angle, in degrees or radians*
ω = *camshaft angular velocity, rad/sec*
ω_d = *damped circular natural frequency, rad/sec*
ω_f = *forcing frequency, rad/sec*
ω_n = *undamped circular natural frequency, rad/sec*
x = *follower displacement, length units*
$\dot{x} = v$ = *follower velocity, length/sec*
$\ddot{x} = a$ = *follower acceleration, length/sec*^{2}
ζ = *damping ratio*

15.1 DYNAMIC FORCE ANALYSIS OF THE FORCE-CLOSED CAM-FOLLOWER

Figure 15-1a shows a simple plate or disk cam driving a spring-loaded, roller follower. This is a force-closed system which depends on the spring force to keep the cam and follower in contact at all times. Figure 15-1b shows a lumped parameter model of this system in which all the **mass** which moves with the follower train is lumped together as m, all the springiness in the system is lumped within the **spring constant** k, and all the **damping** or resistance to movement is lumped together as a damper with coefficient c. The sources of mass which contribute to m are fairly obvious. The masses of the follower stem, the roller, its pivot pin, and any other hardware attached to the moving assembly all add together to create m. Figure 15-1c shows the free-body diagram of the system acted upon by the cam force F_c, the spring force F_s, and the damping force F_d. There will of course also be the effects of mass times acceleration on the system.

Undamped Response

Figure 15-2 shows an even simpler lumped parameter model of the same system as in Figure 15-1 but which omits the damping altogether. This is referred to as a *conservative model* since it conserves energy with no losses. This is not a realistic or safe assumption in this case but will serve a purpose in the path to a better model which will include the damping. The free-body diagram for this mass-spring model is shown in Figure 15-2c. We can write Newton's equation for this one-*DOF* system:



(a) Physical system (b) Lumped model (c) Free-body diagram

FIGURE 15-1

One-*DOF* lumped parameter model of a cam-follower system including damping

$$\sum F = ma = m\ddot{x}$$

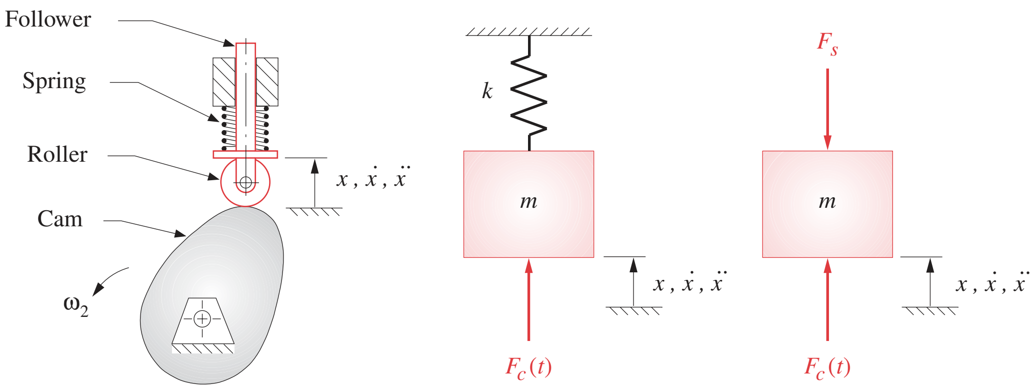
$$F_c(t) - F_s = m\ddot{x}$$

From equation 10.16:

$$F_s = kx$$

then:

$$m\ddot{x} + kx = F_c(t) \tag{15.1a}$$



(a) Physical system (b) Lumped model (c) Free-body diagram

FIGURE 15-2

One-DOF lumped parameter model of a cam-follower system without damping

This is a second-order ordinary differential equation (ODE) with constant coefficients. The complete solution will consist of the sum of two parts, the transient (homogeneous) and the steady state (particular). The homogeneous ODE is

$$
\begin{aligned}
m\ddot{x} + kx &= 0 \\
\ddot{x} &= -\frac{k}{m}x
\end{aligned}
\tag{15.1b}
$$

which has the well-known solution

$$x = A\cos\omega t + B\sin\omega t \tag{15.1c}$$

where A and B are the constants of integration to be determined by the initial conditions. To check the solution, differentiate it twice, assuming constant ω, and substitute in the homogeneous ODE.

$$-\omega^2\left(A\cos\omega t + B\sin\omega t\right) = -\frac{k}{m}\left(A\cos\omega t + B\sin\omega t\right)$$

This is a solution provided that:

$$\omega^2 = \frac{k}{m} \qquad\qquad \omega_n = \sqrt{\frac{k}{m}} \tag{15.1d}$$

The quantity ω_n (rad/sec) is called the *circular natural frequency* of the system and is the frequency at which the system wants to vibrate if left to its own devices. This represents the *undamped natural frequency* since we ignored damping. The *damped natural frequency* will be slightly lower than this value. Note that ω_n is a function only of the physical parameters of the system m and k; thus it is completely determined and unchanging with time once the system is built. By creating a one-*DOF* model of the system, we have limited ourselves to one natural frequency which is an "average" natural frequency usually close to the lowest, or fundamental, frequency.

Any real physical system will also have higher natural frequencies which in general will not be integer multiples of the fundamental. In order to find them we need to create a multi-degree-of-freedom model of the system. The fundamental tone at which a bell rings when struck is its natural frequency defined by this expression. The bell also has overtones which are its other, higher, natural frequencies. The fundamental frequency tends to dominate the transient response of the system.[1]

The circular natural frequency ω_n (rad/sec) can be converted to cycles per second (hertz) by noting that there are 2π radians per revolution and one revolution per cycle:

$$f_n = \frac{1}{2\pi}\omega_n \quad \text{hertz} \tag{15.1e}$$

The constants of integration, A and B in equation 15.1c, depend on the initial conditions. A general case can be stated as

When $t = 0$, let $x = x_0$ and $v = v_0$, where x_0 and v_0 are constants

which gives a general solution to the homogeneous ODE 15.1b of:

$$x = x_0\cos\omega_n t + \frac{v_0}{\omega_n}\sin\omega_n t \tag{15.1f}$$

Equation 15.1f can be put into polar form by computing the magnitude and phase angle:

$$X_0 = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2} \qquad \phi = \arctan\left(\frac{v_0}{x_0\omega_n}\right)$$

then:

$$x = X_0 \cos\left(\omega_n t - \phi\right) \tag{15.1g}$$

Note that this is a pure harmonic function whose amplitude X_0 and phase angle ϕ are a function of the initial conditions and the natural frequency of the system. It will oscillate forever in response to a single, transitory input if there is truly no damping present.

Damped Response

If we now reintroduce the damping of the model in Figure 15-1b and draw the free-body diagram as shown in Figure 15-1c, the summation of forces becomes:

$$F_c\left(t\right) - F_d - F_s = m\ddot{x} \tag{15.2a}$$

Substituting equations 10.16 and 10.17c:

$$m\ddot{x} + c\,\dot{x} + k\,x = F_c\left(t\right) \tag{15.2b}$$

Homogeneous Solution We again separate this differential equation into its homogeneous and particular components. The homogeneous part is:

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0 \tag{15.2c}$$

The solution to this ODE is of the form:

$$x = Re^{st} \tag{15.2d}$$

where R and s are constants. Differentiating versus time:

$$\dot{x} = Rse^{st}$$
$$\ddot{x} = Rs^2e^{st}$$

and substituting in equation 15.2c:

$$Rs^2e^{st} + \frac{c}{m}Rse^{st} + \frac{k}{m}Re^{st} = 0$$

$$\left(s^2 + \frac{c}{m}s + \frac{k}{m}\right)Re^{st} = 0 \tag{15.2e}$$

For this solution to be valid either R or the expression in parentheses must be zero as e^{st} is never zero. If R were zero, then the assumed solution, in equation 15.2d, would also be zero and thus not be a solution. Therefore, the quadratic equation in parentheses must be zero.

$$\left(s^2 + \frac{c}{m}s + \frac{k}{m}\right) = 0 \tag{15.2f}$$

This is called the characteristic equation of the ODE and its solution is:

$$s=\frac{-\dfrac{c}{m}\pm\sqrt{\left(\dfrac{c}{m}\right)^2-4\dfrac{k}{m}}}{2}$$

which has the two roots:

$$s_1=-\frac{c}{2m}+\sqrt{\left(\frac{c}{2m}\right)^2-\frac{k}{m}}$$
$$s_2=-\frac{c}{2m}-\sqrt{\left(\frac{c}{2m}\right)^2-\frac{k}{m}} \tag{15.2g}$$

These two roots of the characteristic equation provide two independent terms of the homogeneous solution:

$$x=R_1e^{s_1t}+R_2e^{s_2t} \qquad \text{for} \quad s_1\neq s_2 \tag{15.2h}$$

If $s_1 = s_2$, then another form of solution is needed. The quantity s_1 will equal s_2 when:

$$\sqrt{\left(\frac{c}{2m}\right)^2-\frac{k}{m}}=0 \qquad \text{or:} \qquad \frac{c}{2m}=\sqrt{\frac{k}{m}}$$

and:

$$c=2m\sqrt{\frac{k}{m}}=2m\omega_n=c_c \tag{15.2i}$$

This particular value of c is called the **critical damping** and is labeled c_c. The system will behave in a unique way when critically damped, and the solution must be of the form:

$$x=R_1e^{s_1t}+R_2te^{s_2t} \qquad \text{for} \quad s_1=s_2=-\frac{c}{2m} \tag{15.2j}$$

It will be useful to define a dimensionless ratio called the **damping ratio** ζ which is the actual damping divided by the critical damping.

$$\zeta=\frac{c}{c_c}$$
$$\zeta=\frac{c}{2m\omega_n} \tag{15.3a}$$

and then:

$$\zeta\omega_n=\frac{c}{2m} \tag{15.3b}$$

The damped natural frequency ω_d is slightly less than the undamped natural frequency ω_n and is:

$$\omega_d=\sqrt{\frac{k}{m}-\left(\frac{c}{2m}\right)^2} \tag{15.3c}$$

We can substitute equations 15.1d and 15.3b into equations 15.2g to get an expression for the characteristic equation in terms of dimensionless ratios:

$$s_{1,2} = -\omega_n\zeta \pm \sqrt{(\omega_n\zeta)^2 - \omega_n{}^2}$$
$$s_{1,2} = \omega_n\left(-\zeta \pm \sqrt{\zeta^2 - 1}\right) \tag{15.4a}$$

This shows that the system response is determined by the damping ratio ζ which dictates the value of the discriminant. There are three possible cases:

Case 1:	$\zeta > 1$	Roots real and unequal	
Case 2:	$\zeta = 1$	Roots real and equal	(15.4b)
Case 3:	$\zeta < 1$	Roots complex conjugate	

Let's consider the response of each of these cases separately.

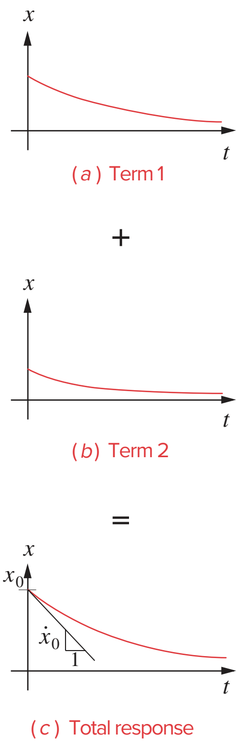


FIGURE 15-3

Transient response of an overdamped system

Case 1: $\zeta > 1$ *overdamped*

The solution is of the form in equation 15.2h and is:

$$x = R_1 e^{\left(-\zeta+\sqrt{\zeta^2-1}\right)\omega_n t} + R_2 e^{\left(-\zeta-\sqrt{\zeta^2-1}\right)\omega_n t} \tag{15.5a}$$

Note that since $\zeta > 1$, both exponents will be negative making x the sum of two decaying exponentials as shown in Figure 15-3. This is the transient response of the system to a disturbance and dies out after a time. There is no oscillation in the output motion. An example of an overdamped system which you have probably encountered is the tone arm on a good-quality record turntable with a "cueing" feature. The tone arm can be lifted up, then released, and it will slowly "float" down to the record. This is achieved by putting a large amount of damping in the system, at the arm pivot. The arm's motion follows an exponential decay curve such as in Figure 15-3.

Case 2: $\zeta = 1$ *critically damped*

The solution is of the form in equation 15.2j and is:

$$x = R_1 e^{-\omega_n t} + R_2 t e^{-\omega_n t} = (R_1 + R_2 t)e^{-\omega_n t} \tag{15.5b}$$

This is the product of a linear function of time and a decaying exponential function and can take several forms depending on the values of the constants of integration, R_1 and R_2, which in turn depend on initial conditions. A typical transient response might look like Figure 15-4. This is the transient response of the system to a disturbance, which response dies out after a time. There is fast response but no oscillation in the output motion. An example of a critically damped system is the suspension system of a new sports car in which the damping is usually made close to critical in order to provide crisp handling response without either oscillating or being slow to respond. A critically damped system will, when disturbed, return to its original position within one bounce. It may overshoot but will not oscillate and will not be sluggish.

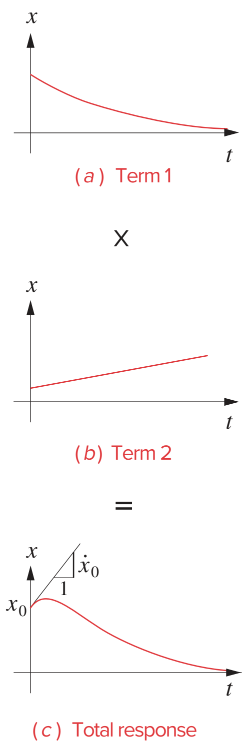


FIGURE 15-4

Transient response of a critically damped system

CASE 3: $\zeta < 1$ *underdamped*

The solution is of the form in equation 15.2h and s_1, s_2 are complex conjugate. Equation 15.4a can be rewritten in a more convenient form as:

$$s_{1,2} = \omega_n\left(-\zeta \pm j\sqrt{1-\zeta^2}\right) \qquad j = \sqrt{-1} \tag{15.5c}$$

Substituting in equation 15.2h:

$$x = R_1 e^{\left(-\zeta + j\sqrt{1-\zeta^2}\right)\omega_n t} + R_2 e^{\left(-\zeta - j\sqrt{1-\zeta^2}\right)\omega_n t}$$

and noting that: $\qquad y^{a+b} = y^a y^b$

$$x = R_1\left[e^{-\zeta\omega_n t} e^{\left(j\sqrt{1-\zeta^2}\right)\omega_n t}\right] + R_2\left[e^{-\zeta\omega_n t} e^{\left(-j\sqrt{1-\zeta^2}\right)\omega_n t}\right]$$

factor:

$$x = e^{-\zeta\omega_n t}\left[R_1 e^{\left(j\sqrt{1-\zeta^2}\right)\omega_n t} + R_2 e^{\left(-j\sqrt{1-\zeta^2}\right)\omega_n t}\right] \tag{15.5d}$$

Substitute the Euler identity from equation 4.4a:

$$x = e^{-\zeta\omega_n t}\left\{\begin{aligned} &R_1\left[\cos\left(\sqrt{1-\zeta^2}\,\omega_n t\right) + j\sin\left(\sqrt{1-\zeta^2}\,\omega_n t\right)\right] \\ &\qquad + R_2\left[\cos\left(\sqrt{1-\zeta^2}\,\omega_n t\right) - j\sin\left(\sqrt{1-\zeta^2}\,\omega_n t\right)\right]\end{aligned}\right\}$$

and simplify: (15.5e)

$$x = e^{-\zeta\omega_n t}\left\{\left(R_1 + R_2\right)\left[\cos\left(\sqrt{1-\zeta^2}\,\omega_n t\right) + \left(R_1 - R_2\right) j\sin\left(\sqrt{1-\zeta^2}\,\omega_n t\right)\right]\right\}$$

Note that R_1 and R_2 are just constants yet to be determined from the initial conditions, so their sum and difference can be denoted as some other constants:

$$x = e^{-\zeta\omega_n t}\left\{A\left[\cos\left(\sqrt{1-\zeta^2}\,\omega_n t\right) + B\sin\left(\sqrt{1-\zeta^2}\,\omega_n t\right)\right]\right\} \tag{15.5f}$$

We can put this in polar form by defining the magnitude and phase angle as:

$$X_0 = \sqrt{A^2 + B^2} \qquad \phi = \arctan\frac{B}{A} \tag{15.5g}$$

then:

$$x = X_0 e^{-\zeta\omega_n t}\cos\left[\left(\sqrt{1-\zeta^2}\,\omega_n t\right) - \phi\right] \tag{15.5h}$$

This is the product of a harmonic function of time and a decaying exponential function where X_0 and ϕ

Figure 15-5 shows the transient response for this **underdamped case**. The response overshoots and oscillates before finally settling down to its final position. Note that if the damping ratio ζ is zero, equation 15.5g reduces to equation 15.1g which is a pure harmonic.

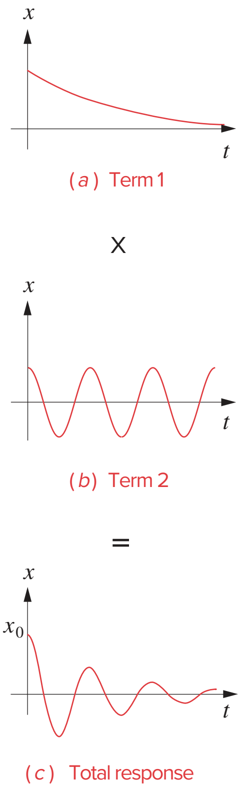


FIGURE 15-5
Transient response of an underdamped system

An example of an underdamped system is a diving board which continues to oscillate after the diver has jumped off, finally settling back to zero position. Many real systems in machinery are underdamped, including the typical cam-follower system. This often leads to vibration problems. It is not usually a good solution simply to add damping to the system as this causes heating and is very energy inefficient. It is better to design the system to avoid the vibration problems.

Particular Solution Unlike the homogeneous solution which is always the same regardless of the input, the particular solution to equation 15.2b will depend on the forcing function $F_c(t)$ which is applied to the cam-follower from the cam. In general the output displacement x of the follower will be a function of similar shape to the input function but will lag the input function by some phase angle. It is quite reasonable to use a sinusoidal function as an example since any periodic function can be represented as a Fourier series of sine and cosine terms of different frequencies (see equations 13.2, 13.3, and their footnote).

Assume the forcing function to be:

$$F_c(t) = F_0 \sin \omega_f t \tag{15.6a}$$

where F_0 is the amplitude of the force and ω_f is its circular frequency. Note that ω_f is unrelated to ω_n or ω_d and may be any value. The system equation then becomes:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega_f t \tag{15.6b}$$

The solution must be of harmonic form to match this forcing function, and we can try the same form of solution as used for the homogeneous solution.

$$x_f(t) = X_f \sin\left(\omega_f t - \psi\right) \tag{15.6c}$$

where:

X_f = amplitude

ψ = phase angle between applied force and displacement

ω_f = angular velocity of forcing function

The factors X_f and ψ are not constants of integration here. They are constants determined by the physical characteristics of the system and the forcing function's frequency and magnitude. They have nothing to do with the initial conditions. To find their values, differentiate the assumed solution twice, substitute in the ODE, and get:

$$X_f = \frac{F_0}{\sqrt{\left(k - m\omega_f^2\right)^2 + \left(c\omega_f\right)^2}}$$

$$\psi = \arctan\left[\frac{c\omega_f}{\left(k - m\omega_f^2\right)^2}\right] \tag{15.6d}$$

Substitute equations 15.1d, 15.2i, and 15.3a and put in dimensionless form:

$$\frac{X_f}{F_0/k} = \frac{1}{\sqrt{\left[1-\left(\frac{\omega_f}{\omega_n}\right)^2\right]^2 + \left(2\zeta\frac{\omega_f}{\omega_n}\right)^2}}$$

$$\psi = \arctan\left[\frac{2\zeta\frac{\omega_f}{\omega_n}}{1-\left(\frac{\omega_f}{\omega_n}\right)^2}\right] \tag{15.6e}$$

The ratio ω_f / ω_n is called the **frequency ratio**. Dividing X_f by the static deflection F_0 / k creates the **amplitude ratio** which defines the relative dynamic displacement compared to the static.

Complete Response The complete solution to our system differential equation for a sinusoidal forcing function is the sum of the homogeneous and particular solutions:

$$x = X_0 e^{-\zeta\omega_n t}\cos\left[\left(\sqrt{1-\zeta^2}\,\omega_n t\right)-\phi\right] + X_f \sin\left(\omega_f t - \psi\right) \tag{15.7}$$

The homogeneous term represents the **transient response** of the system which will die out in time but is reintroduced any time the system is again disturbed. The **particular term** represents the **forced response** or **steady-state response** to a sinusoidal forcing function which will continue as long as the forcing function is present.

Note that the solution to this equation, shown in equations 15.5 and 15.6, depends only on two ratios, the damping ratio ζ which relates the actual damping relative to the critical damping, and the *frequency ratio* ω_f / ω_n which relates the forcing frequency to the natural frequency of the system. Koster[1] found that a typical value for the damping ratio in cam-follower systems is $\zeta = 0.06$, so they are underdamped and can **resonate** if operated at frequency ratios close to 1.

The initial conditions for the specific problem are applied to equation 15.7 to determine the values of X_0 and ϕ. Note that these constants of integration are contained within the homogeneous part of the solution.

15.2 RESONANCE

The natural frequency (and its overtones) are of great interest to the designer as they define the frequencies at which the system will **resonate**. The single-*DOF* lumped parameter systems shown in Figures 15-1 and 15-2 are the simplest possible to describe a dynamic system, yet they contain all the basic dynamic elements. Masses and springs are energy storage elements. A mass stores kinetic energy, and a spring stores potential energy. The damper is a dissipative element. It uses energy and converts it to heat. Thus all the losses in the model of Figure 15-1 occur through the damper.

These are "pure" idealized elements which possess only their own characteristics. That is, the spring has no damping and the damper no springiness, etc. Any system that contains more than one energy storage device such as a mass and a spring will possess at least one natural frequency. If we excite the system at its natural frequency, we will set up the condition called resonance in which the energy stored in the system's elements will oscillate from one element to the other at that frequency. The result can be violent oscillations in the displacements of the movable elements in the system as the energy moves from potential to kinetic form and vice versa.

Figures 15-6a and b show the amplitude and phase angle, respectively, of the displacement response X of the system to a sinusoidal input forcing function at various frequencies ω_f. The forcing frequency ω_f is the angular velocity of the cam. These plots normalize the forcing frequency as a frequency ratio ω_f / ω_n. The amplitude X is normalized by dividing the dynamic deflection x by the static deflection F_0 / k that the same force amplitude would create on the system. Thus at a frequency of zero, the output is

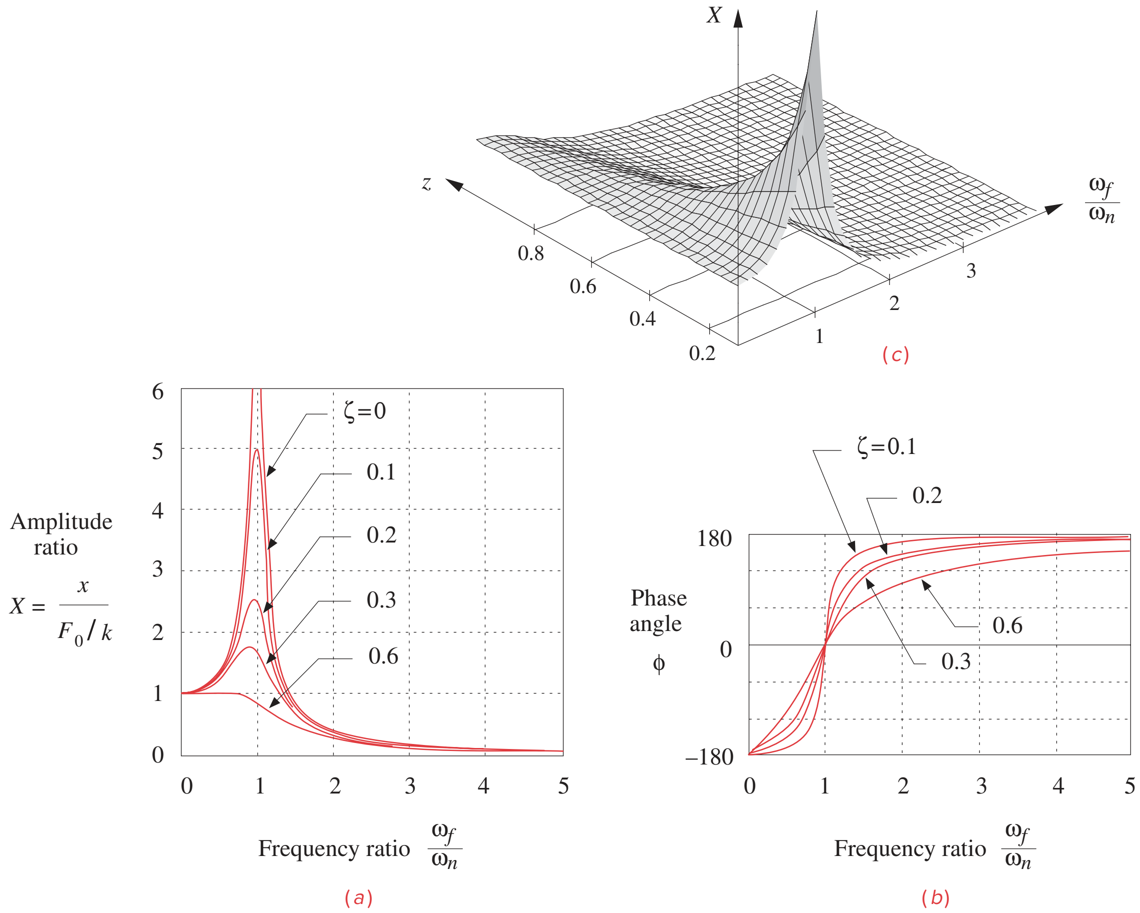


FIGURE 15-6

Amplitude ratio and phase angle of system response one, equal to the static deflection of the spring at the amplitude of the input force. As the forcing frequency increases toward the natural frequency ω_n, the amplitude of the output motion, for zero damping, increases rapidly and becomes theoretically infinite when $\omega_f = \omega_n$. Beyond this point the amplitude decreases rapidly and asymptotically toward zero at high frequency ratios.

The effects of damping ratio ζ can best be seen in Figure 15-6c, which shows a 3-D plot of forced vibration amplitude as a function of both frequency ratio ω_f / ω_n and damping ratio ζ. The addition of damping reduces the amplitude of vibration at the natural frequency, but very large damping ratios are needed to keep the output amplitude less than or equal to the input amplitude. About 50 to 60% of critical damping will eliminate the resonance peak. Unfortunately, most cam-follower systems have damping ratios of less than about 10% of critical. At those damping levels, the response at resonance is about five times the static response. This will create unsustainable stresses in most systems if allowed to occur.

It is obvious that we must avoid driving this system at or near its natural frequency. One result of operation of an underdamped cam-follower system near ω_n can be **follower jump**. The system of follower mass and spring can oscillate violently at its natural frequency and leave contact with the cam. When it does reestablish contact, it may do so with severe impact loads that can quickly fail the materials.

The designer has a degree of control over resonance in that the system's mass *m* and stiffness *k* can be tailored to move its natural frequency away from any required operating frequencies. A common rule of thumb is to design the system to have a fundamental natural frequency ω_n at least ten times the highest forcing frequency expected in service, thus keeping all operation well below the resonance point. This is often difficult to achieve in mechanical systems. One tries to achieve the largest ratio ω_n / ω_f possible nevertheless. It is important to adhere to the fundamental law of cam design and use cam programs with finite jerk in order to minimize vibrations in the follower system.

Some thought and observation of equation 15.1d will show that we would like our system members to be both light (low *m*) and stiff (high *k*) to get high values for ω_n. Unfortunately, the lightest materials are seldom also the stiffest. Aluminum is one-third the weight of steel but is also about one-third as stiff. Titanium is about half the weight of steel but also about half as stiff. Some of the exotic composite materials such as carbon fiber/epoxy offer better stiffness-to-weight ratios but their cost is high and processing is difficult. Another job for *Unobtainium 208!*

Note in Figure 15-6 that the amplitude of vibration at large frequency ratios approaches zero. So, if the system can be brought up to speed through the resonance point without damage and then kept operating at a large frequency ratio, the vibration will be minimal. An example of systems designed to be run this way are large devices that must run at higher speed such as electrical power generators. Their large mass creates a lower natural frequency than their required operating speeds. They are "run up" as quickly as possible through the resonance region to avoid damage from their vibrations and "run down" quickly through resonance when stopping them. They also have the advantage of long duty cycles of constant-speed operation in the safe frequency region between infrequent starts and stops.

15.3 KINETOSTATIC FORCE ANALYSIS OF THE FORCE-CLOSED CAM-FOLLOWER

The previous sections introduced **forward dynamic analysis** and the solution to the system differential equation of motion (equation 15.2b). The applied force $F_c(t)$ is presumed to be known, and the system equation is solved for the resulting displacement x from which its derivatives can also be determined. The **inverse dynamics**, or **kinetostatics**, approach provides a quick way to determine how much spring force is needed to keep the follower in contact with the cam at a chosen design speed. The displacement and its derivatives are defined from the kinematic design of the cam based on an assumed constant angular velocity ω of the cam. Equation 15.2b can be solved algebraically for the force $F_c(t)$ in a spring-loaded cam-follower system provided that values for mass m, spring constant k, preload F_{pl}, and damping factor c are known in addition to the displacement, velocity, and acceleration functions.

Figure 15-1a shows a simple plate or disk cam driving a spring-loaded, roller follower. This is a force-closed system which depends on the spring force to keep the cam and follower in contact at all times. Figure 15-1b shows a lumped parameter model of this system in which all the **mass** that moves with the follower train is lumped together as m, all the springiness in the system is lumped within the **spring constant** k, and all the **damping** or resistance to movement is lumped together as a damper with coefficient c.

The designer has a large degree of control over the system spring constant k_{eff} as it tends to be dominated by the k_s of the physical return spring. The elasticities of the follower parts also contribute to the overall system k_{eff} but are usually much stiffer than the physical spring. If the follower stiffness is in series with the return spring, as it often is, equations 10.19 show that the softest spring in series will dominate the effective spring constant. Thus the return spring will virtually determine the overall k unless some parts of the follower train have similarly low stiffness.

The designer will choose or design the return spring and thus can specify both its k and the amount of preload deflection x_0 to be introduced at assembly. Preload of a spring occurs when it is compressed (or extended if an extension spring) from its *free length* to its initial assembled length. This is a necessary and desirable situation as we want some residual force on the follower even when the cam is at its lowest displacement. This will help maintain good contact between the cam and follower at all times. This spring preload $F_{pl} = kx_0$ adds a constant term to equation 15.2b which becomes:

$$F_c(t) = m\ddot{x} + c\dot{x} + kx + F_{pl} \tag{15.8a}$$

or:

$$F_c(t) = m\ddot{x} + c\dot{x} + k(x + x_0) \tag{15.8b}$$

The value of m is determined from the effective mass of the system as lumped in the single-*DOF* model of Figure 15-1. The value of c for most cam-follower systems can be estimated for a first approximation to be about 0.05 to 0.10 of the critical damping c_c as defined in equation 15.2i. Koster[1] found that a typical value for the damping ratio in cam-follower systems is $\zeta = 0.06$.

Calculating the damping c based on an assumed value of ζ requires specifying a value for the overall system k and for its effective mass. The choice of k will affect both the natural frequency of the system for a given mass and the available force to keep the joint closed. Some iteration will probably be needed to find a good compromise. A selection of data for commercially available helical coil springs is provided in Appendix D. Note in equations 15.8 that the terms involving acceleration and velocity can be either positive or negative. The terms involving the spring parameters k and F_{pl} are the only ones that are always positive. So, to keep the overall function always positive requires that the spring force terms be large enough to counteract any negative values in the other terms. Typically, the acceleration is larger numerically than the velocity, so the negative acceleration usually is the principal cause of a negative force F_c.

The principal concern in this analysis is to keep the cam force always positive in sign as its direction is defined in Figure 15-1. The cam force is shown as positive in that figure. In a force-closed system the cam can only push on the follower. It cannot pull. The follower spring is responsible for providing the force needed to keep the joint closed during the negative acceleration portions of the follower motion. The damping force also can contribute, but the spring must supply the bulk of the force to maintain contact between the cam and follower. If the force F_c goes negative at any time in the cycle, the follower and cam will part company, a condition called **follower jump**. When they meet again, it will be with large and potentially damaging impact forces. The follower jump, if any, will occur near the point of maximum negative acceleration. Thus we must select the spring constant and preload to guarantee a positive force at all points in the cycle. In automotive engine valve cam applications, follower jump is also called *valve float*, because the valve (follower) "floats" above the cam, also periodically impacting the cam surface. This will occur if the cam rpm is increased to the point that the larger negative acceleration makes the follower force negative. The "redline" maximum engine rpm often indicated on its tachometer is to warn of impending valve float above that speed which will damage the cam and follower.

Program DYNACAM allows the iteration of equations 15.8 to be done quickly for any cam whose kinematics have been defined in that program. The program's *Dynamics* button will solve equations 15.8 for all values of camshaft angle, using the displacement, velocity, and acceleration functions previously calculated for that cam design in the program. The program requires values for the effective system mass m, effective spring constant k, preload F_{pl}, and the assumed value of the damping ratio ζ. These values need to be determined for the model by the designer using the methods described in Sections 10.11 and 10.12. The calculated force at the cam-follower interface can then be plotted or its values printed in tabular form. The system's natural frequency is also reported when the tabular force data are printed.

EXAMPLE 15-1

Kinetostatic Force Analysis of a Force-Closed (Spring-Loaded) Cam-Follower System.

Given: A translating roller follower as shown in Figure 15-1 is driven by a force-closed radial plate cam which has the following program:

15

Segment 1: Rise 1 inch in 50° with modified sine acceleration
Segment 2: Dwell for 40°
Segment 3: Fall 1 inch in 50° with cycloidal displacement
Segment 4: Dwell for 40°
Segment 5: Rise 1 inch in 50° with 3-4-5 polynomial displacement

Segment 6: Dwell for 40°
Segment 7: Fall 1 inch in 50° with 4-5-6-7 polynomial displacement
Segment 8: Dwell for 40°
Camshaft angular velocity is 18.85 rad/sec.
Follower effective mass is 0.0738 in-lb-sec^2 (blob).
Damping is 15% of critical ($\zeta = 0.15$).

Problem: Find the spring constant and spring preload to maintain contact between the cam and follower and calculate the dynamic force function for the cam. Find the system natural frequency with the selected spring. Keep the pressure angle under 30°.

Solution:

1 Calculate the kinematic data (follower displacement, velocity, acceleration, and jerk) for the specified cam functions. The acceleration for this cam is shown in Figure 15-7 and has a maximum value of 3504 in/sec^2. See Chapter 8 to review this procedure.

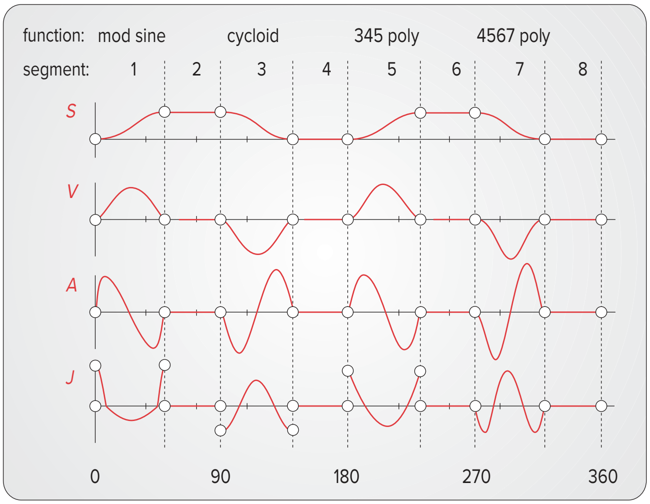
2 Calculate the pressure angle and radius of curvature for trial values of prime circle radius, and size the cam to control these values. Figure 15-8 shows the pressure angle function and Figure 15-9 the radii of curvature for this cam with a prime circle radius of 4 in and zero eccentricity. The maximum pressure angle is 29.2° and the minimum radius of curvature is 1.7 in. Figure 8-50 shows the finished cam profile. See Chapter 8 to review these calculations.

3 With the kinematics of the cam defined, we can address its dynamics. To solve equations 15.8 for cam force, we must assume values for the spring constant k and the preload F_{pl}. The value of c can be calculated from equation 15.3a using the given mass m, the damping factor ζ, and assumed k. The kinematic parameters are known.

4 Program DYNACAM does this computation for you. The dynamic force that results from an assumed k of 150 lb/in and a preload of 75 lb is shown in Figure 15-10a. The damping coef-

Segment Number	Function Used	Start Angle	End Angle	Delta Angle
1	ModSine rise	0	50	50
2	Dwell	50	90	40
3	Cycloid fall	90	140	50
4	Dwell	140	180	40
5	345 poly rise	180	230	50
6	Dwell	230	270	40
7	4567 poly fall	270	320	50
8	Dwell	320	360	40

(*a*) Cam program specifications



(*b*) Plots of cam-follower's *S V A J* diagrams

FIGURE 15-7

S V A J diagrams for Examples 15-1 and 15-2

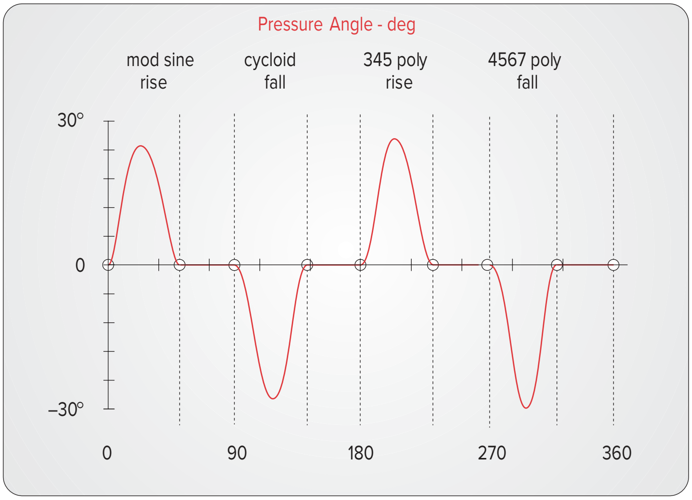


FIGURE 15-8

Pressure angle plot for Examples 15-1 and 15-2

ficient $c = 0.998$. Note that the force dips below the zero axis in two places during negative acceleration. These are locations of follower jump. The follower has left the cam during the fall because the spring does not have enough available force to keep the follower in contact with the rapidly falling cam. Open the file E15-01.cam in DYNACAM and provide the specified k and F_{pl} to see this example. Another iteration is needed to improve the design.

5 Figure 15-10b shows the dynamic force for the same cam with a spring constant of $k = 200$ lb/

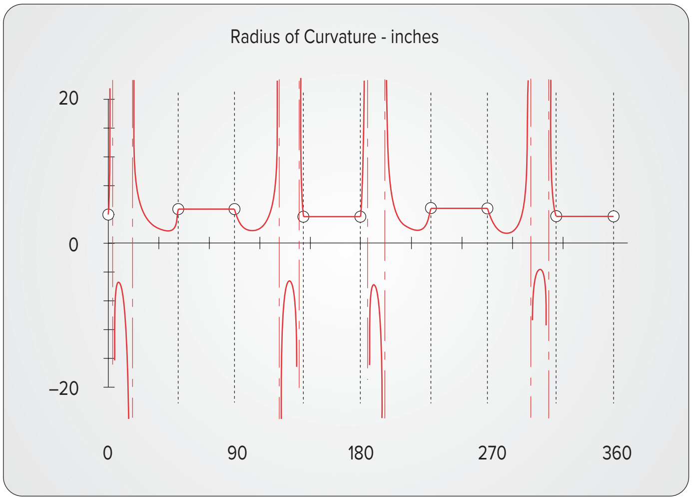
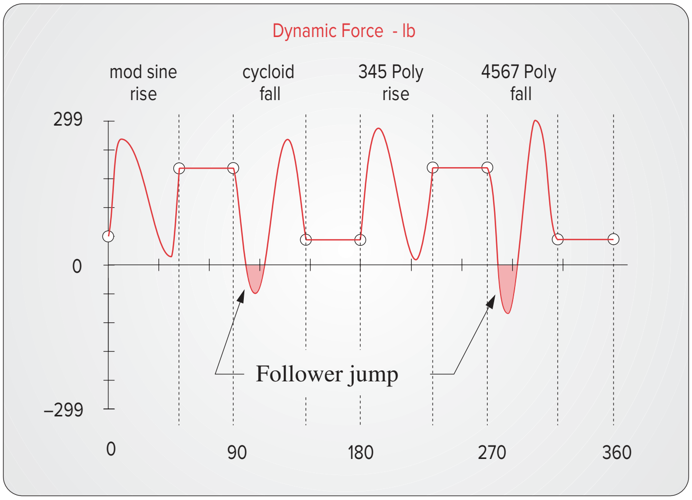


FIGURE 15-9

Radius of curvature of a four-dwell cam for Examples 15-1 and 15-2

(a) Insufficient spring force allows follower jump



(b) Sufficient spring force keeps the dynamic force positive

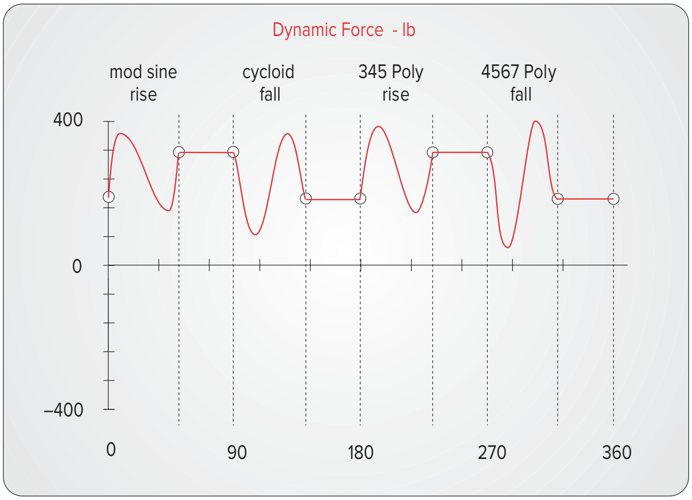


FIGURE 15-10

Dynamic forces in a force-closed cam-follower system

in and a preload of 150 lb. The damping coefficient $c = 1.153$. This additional force has lifted the function up sufficiently to keep it positive everywhere. There is no follower jump in this case. The maximum force during the cycle is 400.4 lb. A margin of safety has been provided by keeping the minimum force comfortably above the zero line at 36.9 lb. Run example #5 in the program and provide the specified spring constant and preload values to see this example.

6 The undamped and damped fundamental natural frequencies can be calculated for the system from equations 15.1d and 15.3c, respectively, and are:

$$\omega_n = 52.06 \text{ rad/sec}; \qquad \omega_d = 51.98 \text{ rad/sec}$$

15

15.4 KINETOSTATIC FORCE ANALYSIS OF THE FORM-CLOSED CAM-FOLLOWER

Section 8.1 described two types of joint closure used in cam-follower systems, **force closure** and **form closure**. Force closure uses an open joint and requires a spring or other force source to maintain contact between the elements. Form closure provides a geometric constraint at the joint such as the cam groove shown in Figure 15-11a or the conjugate cams of Figure 15-11b. No spring is needed to keep the follower in contact with these cams. The follower will run against one side or the other of the groove or conjugate pair as necessary to provide both positive and negative forces. Since there is no spring in this system, its dynamic force equation 15.8 simplifies to:

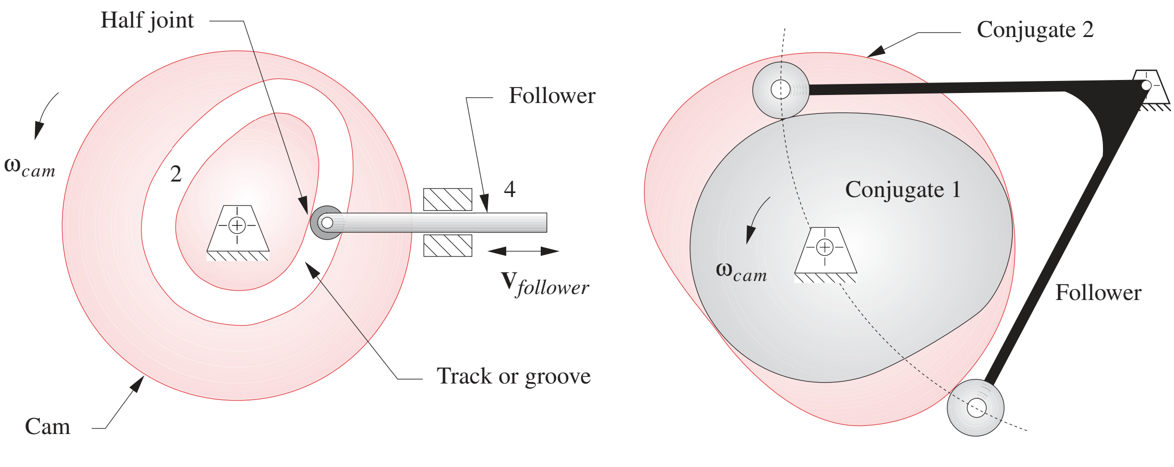
$$F_c(t) = m\ddot{x} + c\,\dot{x} \tag{15.9}$$

Note that there is now only one energy storage element in the system (the mass), so, theoretically, resonance is not possible. There is no natural frequency for it to resonate at. This is the chief advantage of a form-closed system over a force-closed one. Follower jump will not occur, short of complete failure of the parts, no matter how fast the system is run. This arrangement is sometimes used in high-performance or racing engine valve trains to allow higher redline engine speeds without valve float. In engine valve trains, a form-closed cam-follower valve train is called a *desmodromic* system.

As with any design, there are trade-offs. While the form-closed system typically allows higher operating speeds than a comparable force-closed system, it is not free of all vibration problems. Even though there is no physical return spring in the system, the follower train, the camshaft, and all other parts still have their own spring constants which abruptly shift from one side of the cam groove to the other. There cannot be zero clearance between the roller follower and the groove and still have it operate. Even if the clearance is very small, there will still be an opportunity for the follower to develop some velocity in its short trip across the groove, and it will impact the other side. Track cams of the type shown in Figure 15-11a typically fail at the points where the acceleration reverses sign, due to many cycles of crossover shock. Note also that the roller follower has to reverse direction every time it crosses over to the other side of the groove. This causes significant follower slip and high wear on the follower compared to an open, force-closed cam where the follower will have less than 1% slip.

Because there are two cam surfaces to machine and because the cam track, or groove, must be cut and ground to high precision to control the clearance, form-closed cams tend to be more expensive to manufacture than force-closed cams. Track cams usually must be ground after heat treatment to correct the distortion of the groove resulting from the high temperatures. Grinding significantly increases cost. Many force-closed cams are not ground after heat treatment and are used as-milled. Though the conjugate cam approach avoids the groove tolerance and heat treat distortion problems, there are still two matched cam surfaces to be made per cam. Thus, the desmodromic cam's dynamic advantages come at a significant cost premium.

We will now repeat the cam design of Example 15-1, modified for desmodromic operation. This is simple to do with program DYNACAM by setting the spring constant and preload values to zero, which assumes that the follower train is a rigid body. A more accurate result can be obtained by calculating and using the effective spring constant of



(*a*) Form-closed cam with translating follower

(*b*) Conjugate cams on common shaft

FIGURE 15-11

Form-closed cam-follower systems

the combination of parts in the follower train, once their geometries and materials are defined. The dynamic forces will now be negative as well as positive, but a form-closed cam can both push and pull.

EXAMPLE 15-2

Dynamic Force Analysis of a Form-Closed (Desmodromic) Cam-Follower System.

Given: A translating roller follower as shown in Figure 15-11a is driven by a form-closed radial plate cam which has the following program:

Segment 1: Rise 1 inch in 50° with modified sine acceleration
Segment 2: Dwell for 40°
Segment 3: Fall 1 inch in 50° with cycloidal displacement
Segment 4: Dwell for 40°
Segment 5: Rise 1 inch in 50° with 3-4-5 polynomial displacement
Segment 6: Dwell for 40°
Segment 7: Fall 1 inch in 50° with 4-5-6-7 polynomial displacement
Segment 8: Dwell for 40°
Camshaft angular velocity is 18.85 rad/sec.
Follower effective mass is 0.0738 in-lb-sec^2 (blob).
Damping is 15% of critical ($\zeta = 0.15$).

Problem: Compute the dynamic force function for the cam. Keep the pressure angle < 30°.

Solution:

1 Calculate the kinematic data (follower displacement, velocity, acceleration, and jerk) for the specified cam functions. The acceleration for this cam is shown in Figure 15-7 and has a maximum value of 3504 in/sec^2. See Chapter 8 to review this procedure.

2 Calculate radius of curvature and pressure angle for trial values of prime circle radius, and size the cam to control these values. Figure 15-8 shows the pressure angle function and Figure 15-9 the radii of curvature for this cam with a prime circle radius of 4 in and zero eccentricity. The maximum pressure angle is 29.2° and the minimum radius of curvature is 1.7 in. Figure 8-50 shows the finished cam profile. See Chapter 8 to review these calculations.

3 With the kinematics of the cam defined, we can address its dynamics. To solve equation 15.9 for the cam force, we assume zero values for the spring constant k and the preload F_{pl}. The value of c is assumed to be the same as in Example 15-1, i.e., 1.153. The kinematic parameters are known.

4 Program DYNACAM does this computation for you. The dynamic force that results is shown in Figure 15-12. Note that the force is now more nearly symmetric about the axis and its peak absolute value is 289 lb. Crossover shock will occur each time the follower force changes sign. Open the file E15-02.cam in DYNACAM to see this example.

Compare the dynamic force plots for the force-closed system (Figure 15-10b, and the form-closed system (Figure 15-12). The absolute peak force magnitude on either side of the track in the form-closed cam is less than that on the spring-loaded one. This shows the penalty that the spring imposes on the system in order to keep the joint closed. Thus, either side of the cam groove will experience lower stresses than will the open cam, except for the areas of crossover shock.

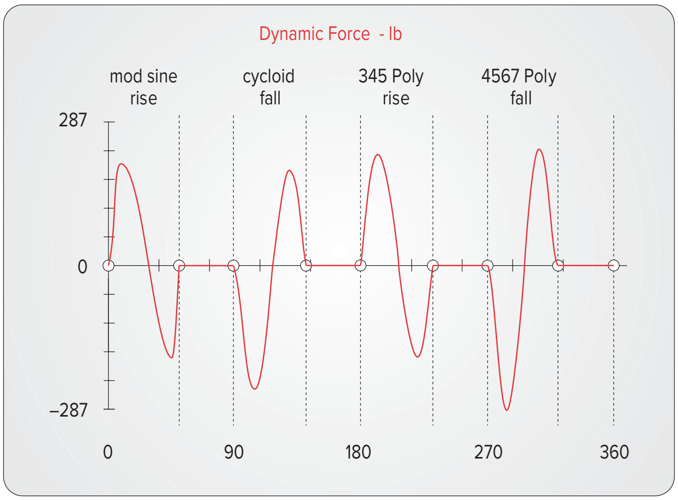


FIGURE 15-12

Dynamic force in a form-closed cam-follower system

15.5 KINETOSTATIC CAMSHAFT TORQUE

The kinetostatic analysis assumes that the camshaft will operate at some constant speed ω. As we saw in the case of the fourbar linkage in Chapter 11 and with the slider-crank mechanism in Chapter 13, the input torque must vary over the cycle if the shaft velocity remains constant. The torque is calculated from the power relationship, ignoring losses.

$$\text{Power in} = \text{Power out}$$
$$T_c\omega = F_f V$$
$$T_c = \frac{F_f V}{\omega} = \frac{\left(F_c \cos\phi\right) V}{\omega} \tag{15.10}$$

where T_c is camshaft torque, ω is camshaft angular velocity, F_c is force between the cam and follower along the common normal, and F_f is the component of F_c in the direction of follower velocity V as defined by the pressure angle ϕ.

Once the cam force F_c has been calculated from either equation 15.8 or 15.9, T_c is easily found since V, ϕ, and ω are known for all values of cam angle θ. Figure 15-13a shows the camshaft input torque needed to drive the force-closed cam designed in Example 15-1. Figure 15-13b shows the camshaft input torque needed to drive the form-closed cam designed in Example 15-2. Note that the torque required to drive the force-closed (spring-loaded) system is significantly higher than that needed to drive the form-closed (track) cam. The spring force is also extracting a penalty here as energy must be stored in the spring during the rise portions which will tend to slow the camshaft. This stored energy is then returned to the camshaft during the fall portions, tending to speed it up. The spring loading causes larger oscillations in the torque.

A flywheel can be sized and fitted to the camshaft to smooth these variations in torque just as was done for the fourbar linkage in Section 11.11. See that section for the design procedure. Program DYNACAM integrates the camshaft torque function pulse by pulse and prints those areas to the screen. These energy data can be used to calculate the required flywheel size for any selected coefficient of fluctuation.

One useful way to compare alternate cam designs is to look at the torque function as well as at the dynamic force. A smaller torque variation will require a smaller motor and/or flywheel and will run more smoothly. Three different designs for a single-dwell cam were explored in Chapter 8. (See Examples 8-6, 8-7, and 8-8.) All had the same lift and duration but used different cam functions. One was a double harmonic, one cycloidal, and one a sixth-degree polynomial. On the basis of their kinematic results, principally acceleration magnitude, we found that the polynomial design was superior. We will now revisit this cam as an example and compare its dynamic force and torque using the same three cam functions.

EXAMPLE 15-3

Compare Dynamic Torques and Forces Among Three Alternate Designs of the Same Cam.

Given: A translating roller follower as shown in Figure 15-1 is driven by a force-closed radial plate cam which has the following program:

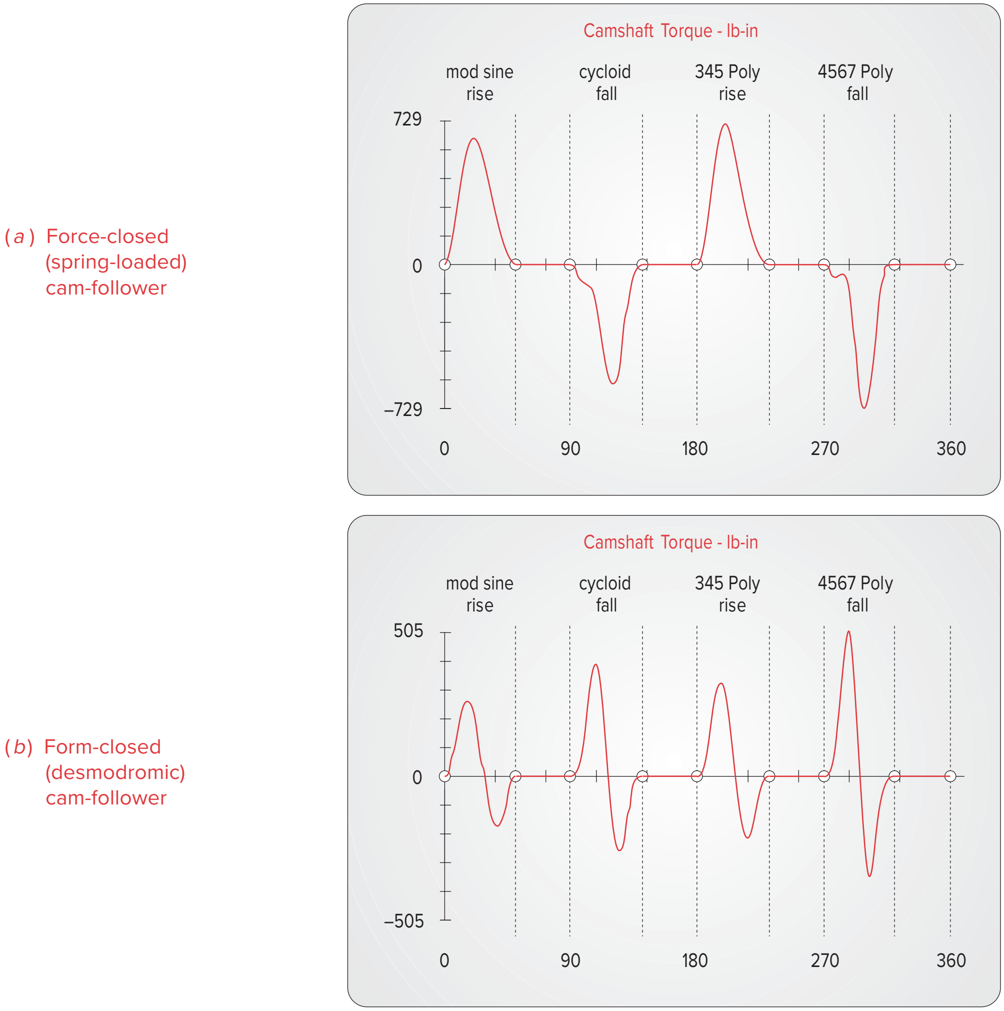


FIGURE 15-13

Input torque in force- and form-closed cam-follower systems

Design 1

Segment 1: Rise 1 inch in 90° double harmonic displacement
Segment 2: Fall 1 inch in 90° double harmonic displacement
Segment 3: Dwell for 180°

Design 2

Segment 1: Rise 1 inch in 90° cycloidal displacement
Segment 2: Fall 1 inch in 90° cycloidal displacement
Segment 3: Dwell for 180°

Design 3

Segment 1: Rise 1 inch in 90° and fall 1 inch in 90° with polynomial displacement. (A single polynomial can create both rise and fall.)
Segment 2: Dwell for 180°

Camshaft angular velocity is 15 rad/sec. Follower effective mass is 0.0738 in-lb-sec^2 (or blob). Damping is 15% of critical ($\zeta = 0.15$).

Find: The dynamic force and torque functions for the cam. Compare their peak magnitudes for the same prime circle radius.

Solution: Note that these are the same kinematic cam designs as are shown in Figures 8-27, 8-28, and 8-30.

1 Calculate the kinematic data (follower displacement, velocity, acceleration, and jerk) for each of the specified cam designs. See Chapter 8 to review this procedure.

2 Calculate the radius of curvature and pressure angle for trial values of prime circle radius, and size the cam to control these values. A prime circle radius of 3 in gives acceptable pressure angles and radii of curvature. See Chapter 8 to review these calculations.

3 With the kinematics of the cam defined, we can address its dynamics. To solve equation 15.1a for the cam force, we will assume a value of 50 lb/in for the spring constant k and adjust the preload F_{pl} for each design to obtain a minimum dynamic force of about 10 lb. For design 1 this requires a spring preload of 28 lb; for design 2, 15 lb; and for design 3, 10 lb.

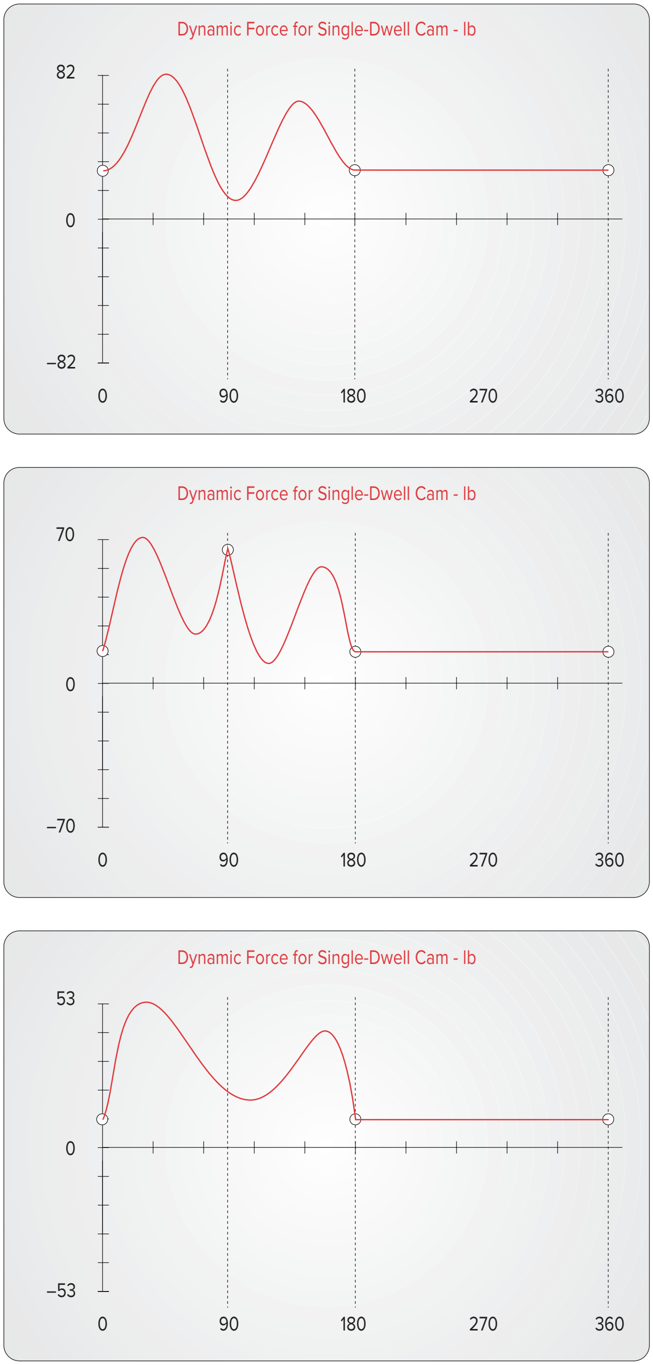
4 The value of damping c is calculated from equation 15.2i. The kinematic parameters x, v, and a are known from the prior analysis.

5 Program DYNACAM will do these computations for you. The dynamic forces that result from each design are shown in Figure 15-14 and the torques in Figure 15-15. Note that the force is largest for design 1 at 82-lb peak and least for design 3 at 53-lb peak. The same ranking holds for the torques which range from 96 lb-in for design 1 to 52 lb-in for design 3. These represent reductions of 35% and 46% in the dynamic loading due to a change in the kinematic design. Not surprisingly, the sixth-degree polynomial design which had the lowest acceleration also has the lowest forces and torques and is the clear winner. Open the files E08-06.cam, E08-07.cam, and E08-08.cam in program DYNACAM to see these results.

15.6 MEASURING DYNAMIC FORCES AND ACCELERATIONS

As described in previous sections, cam-follower systems tend to be underdamped. This allows significant oscillations and vibrations to occur in the follower train. Dynamic forces and accelerations can be measured fairly easily in operating machinery. Compact, piezoelectric force and acceleration transducers are available that have frequency response ranges in the high thousands of hertz. Strain gages provide strain measurements that are proportional to force and have bandwidths of a kilohertz or better.

Figure 15-16 shows acceleration and force curves as measured on the follower train of a single overhead camshaft (SOHC) valve train in a 1.8-liter four-cylinder inline engine.[2] The nonfiring engine was driven by an electric motor on a dynamometer. The camshaft is turning at 500, 2000, and 3000 rpm (1000, 4000, and 6000 crankshaft rpm), respectively, in the three plots of Figure 15-16a, b, and c. Acceleration was measured with a piezoelectric accelerometer attached to the head of one intake valve, and the force was calculated from the output of strain gages placed on the rocker arm for that intake valve. The theoretical follower acceleration curve (as designed) is superposed on the measured



(*a*) Double harmonic rise - double harmonic fall

(*b*) Cycloidal rise - cycloidal fall

(*c*) Sixth-degree polynomial

FIGURE 15-14

Dynamic forces in three different designs of a single-dwell cam (*a*) Double harmonic rise - double harmonic fall

(*b*) Cycloidal rise - cycloidal fall

(*c*) Sixth-degree polynomial

FIGURE 15-15

Dynamic input torque in three different designs of a single-dwell cam (*a*) 500 rpm

(*b*) 2000 rpm

(*c*) 3000 rpm

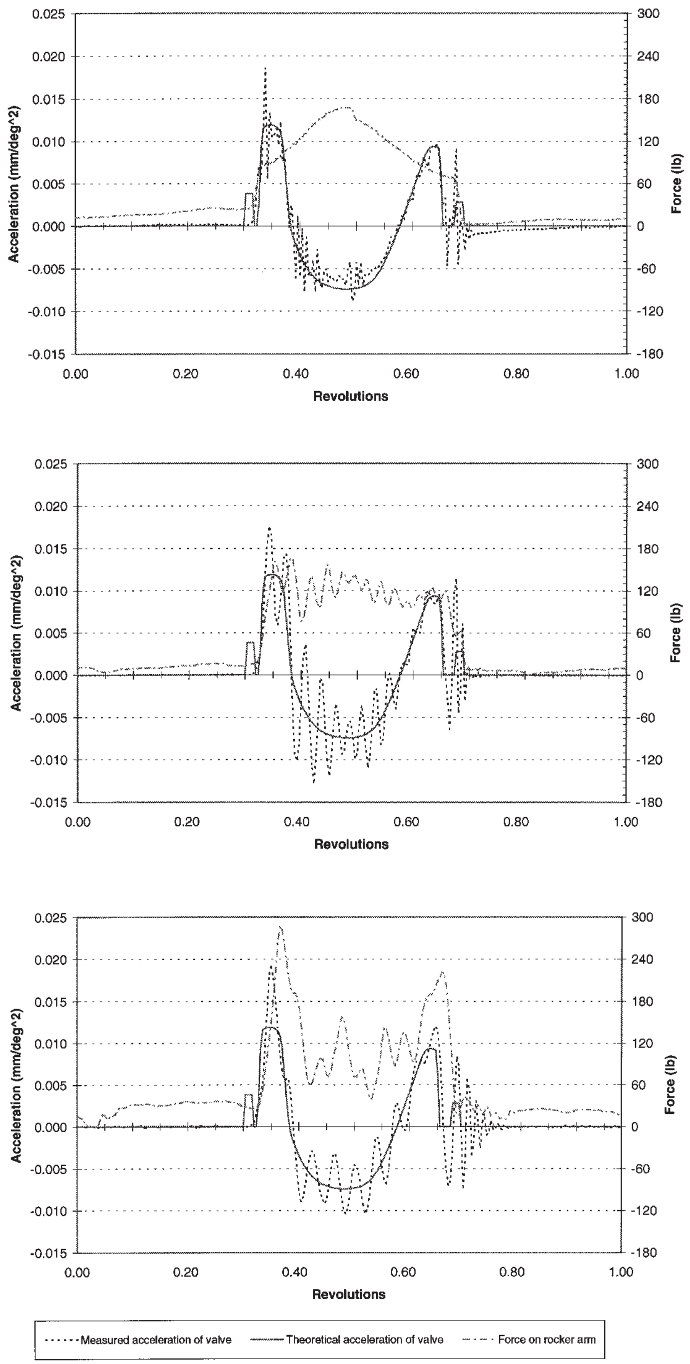


FIGURE 15-16

Valve acceleration and rocker arm force in a single-overhead-cam valve train acceleration curve. All acceleration measurements are converted to units of mm/deg^2 (i.e., normalized to camshaft speed) to allow comparison with one another and with the theoretical acceleration curve.

At 500 camshaft rpm, the measured acceleration closely matches the theoretical acceleration curve with some minor oscillations due to spring vibration. At 2000 camshaft rpm, significant oscillation in the measured acceleration is seen during the first positive and in the negative acceleration phase. This is due to the valve spring vibrating at its natural frequency in response to excitation by the cam. This is called "spring surge" and is a significant factor in valve spring fatigue failure. At 3000 camshaft rpm, the spring surge is still present but is less prominent as a percentage of total acceleration. The frequency content of the cam's forcing function passed through the first natural frequency of the valve spring at about 2000 camshaft rpm, causing the spring to resonate. The same effects can be seen in the rocker arm force. Everything in a machine tends to sympathetically vibrate at its own natural frequency when excited by any input forcing function. Sensitive transducers such as accelerometers will pick up these vibrations as they are transmitted through the structure.

15.7 PRACTICAL CONSIDERATIONS

Koster[1] proposes some general rules for the design of cam-follower systems for high-speed operation based on his extensive dynamic modeling and experimentation.

To minimize the positional error and residual acceleration:

1. Keep the total lift of the follower to a minimum.
2. If possible, arrange the follower spring to preload all pivots in a consistent direction to control backlash in the joints.
3. Keep the duration of rises and falls as long as possible.
4. Keep follower train mass low and follower train stiffness high to increase natural frequency.
5. Any lever ratios present will change the effective stiffness of the system by the square of the ratio. Try to keep lever ratios close to 1.
6. Make the camshaft as stiff as possible **both in torsion and in bending**. This is perhaps the most important factor in controlling follower vibration.
7. Reduce pressure angle by increasing the cam pitch circle diameter.
8. Use low backlash or antibacklash gears in the camshaft drive train.

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2 **Norton, R. L., et al.** (1998). "Analyzing Vibrations in an IC Engine Valve Train." SAE Paper: 980570.

TABLE P15-0
Topic/Problem Matrix

15.1 Dynamic Force Analysis and Resonance
15-6, 15-25, 15-27, 15-28, 15-29

15.3 Kinetostatic Force Analysis
15-7, 15-8, 15-9, 15-10, 15-11, 15-12, 15-13, 15-14,15-18, 15-19, 15-20, 15-21,15-22,15-23, 15-24, 15-30

15.5 Camshaft Torque
15-1, 15-2, 15-3, 15-4, 15-5, 15-15, 15-16, 15-17, 15-31, 15-32

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15.10 PROBLEMS

Program DYNACAM *may be used to solve these problems where applicable. Where units are unspecified, work in any consistent units system you wish. Appendix D contains some pages from a catalog of commercially available helical coil springs to aid in designing realistic solutions to these problems. Other spring information can be found on the Web.*

*†‡15-1 Design a double-dwell cam to move a 2-in-dia roller follower of mass = 2.2 bl from 0 to 2.5 inches in 60° with modified sine acceleration, dwell for 120°, fall 2.5 inches in 30° with cycloidal motion, and dwell for the remainder. The total cycle must take 4 sec. Size a return spring and specify its preload to maintain contact between the cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.2 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

*†‡15-2 Design a double-dwell cam to move a 2-in-dia roller follower of mass = 1.4 bl from 0 to 1.5 inches in 45° with 3-4-5 polynomial motion, dwell for 150°, fall 1.5 inches in 90° with 4-5-6-7 polynomial motion, and dwell for the remainder. The total cycle must take 6 sec. Size a return spring and specify its preload to maintain contact between the cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.1 times critical. Repeat for a form-closed cam. Compare the dynamic force,

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM.

*†‡15-3 Design a single-dwell cam to move a 2-in-dia roller follower of mass = 3.2 bl from 0 to 2 inches in 60°, fall 2 inches in 90°, and dwell for the remainder. The total cycle must take 5 sec. Use a seventh-degree polynomial. Size a return spring and specify its preload to maintain contact between the cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.15 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

*†‡15-4 Design a three-dwell cam to move a 2-in-dia roller follower of mass = 0.4 bl from 0 to 2.5 inches in 40°, dwell for 100°, fall 1.5 inches in 90°, dwell for 20°, fall 1 inch in 30°, and dwell for the remainder. The total cycle must take 10 sec. Choose suitable programs for the rise and fall to minimize dynamic forces and torques. Size a return spring and specify its preload to maintain contact between the cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.12 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

*†‡15-5 Design a four-dwell cam to move a 2-in-dia, 1.25 bl mass roller follower from 0 to 2.5 inches in 40°, dwell for 100°, fall 1.5 inches in 90°, dwell for 20°, fall 0.5 inch in 30°, dwell for 40°, fall 0.5 inch in 30°, and dwell for the remainder. The total cycle is 15 sec. Choose suitable programs for the rise and fall to minimize dynamic forces and torques. Size a return spring and specify its preload to maintain contact between the cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.18 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

*†‡15-6 A mass-spring damper system as shown in Figure 15-1b has the values shown in Table P15-1. Find the undamped and damped natural frequencies and the value of critical damping for the system(s) assigned.

†15-7 Figure P15-1 shows a cam-follower system. The dimensions of the solid, rectangular 2 x 2.5 in cross-section aluminum arm are given. The cutout for the 2-in-diameter, 1.5-in-wide steel roller follower is 3 in long. Find the arm's mass, center of gravity location, and mass moment of inertia about both its *CG* and the arm pivot. Create a linear, one-*DOF* lumped mass model of the dynamic system referenced to the cam-follower and calculate the cam-follower force for one revolution. The cam is a pure eccentric with eccentricity = 0.5 in and turns at 500 rpm. The spring has a rate of 123 lb/in and a preload of 173 lb. Ignore damping.

†‡15-8 Repeat Problem 15-7 for a double-dwell cam to move the roller follower from 0 to 2.5 inches in 60° with modified sine acceleration, dwell for 120°, fall 2.5 inches in 30° with cycloidal motion, and dwell for the remainder. Cam speed is 100 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.10.

†‡15-9 Repeat Problem 15-7 for a double-dwell cam to move the roller follower from 0 to 1.5 inches in 45° with 3-4-5 polynomial motion, dwell for 150°, fall 1.5 inches in 90° with 4-5-6-7 polynomial motion, and dwell for the remainder. Cam speed is 250 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.15.

†‡15-10 Repeat Problem 15-7 for a single-dwell cam to move the follower from 0 to 2 inches in 60°, fall 2 inches in 90°, and dwell for the remainder. Use a seventh-degree polynomial. Cam speed is 250 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.15.

TABLE P15-1
Problem 15-6

	m	*k*	*c*
a	1.2	14	1.1
b	2.1	46	2.4
c	30.0	2	0.9
d	4.5	25	3.0
e	2.8	75	7.0
f	12.0	50	14.0

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM.

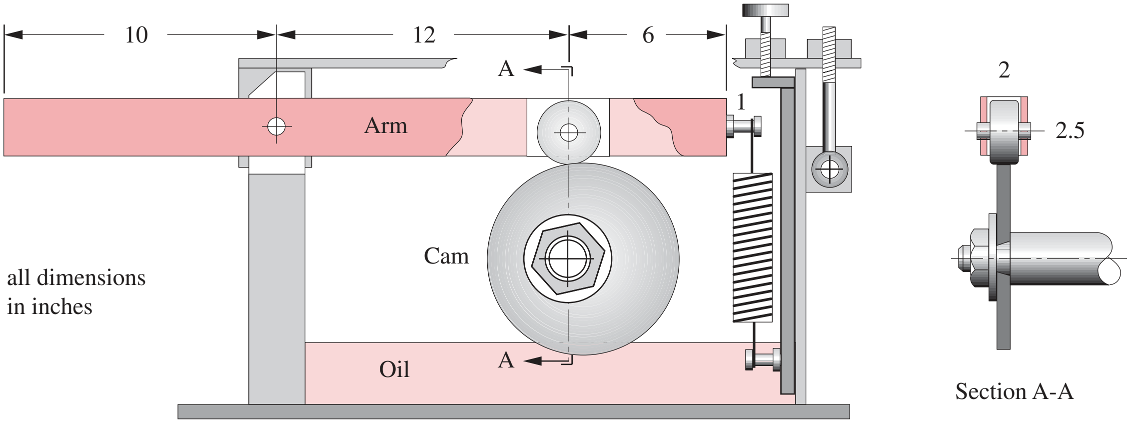

FIGURE P15-1

Problems 15-7 to 15-11 and 15-27

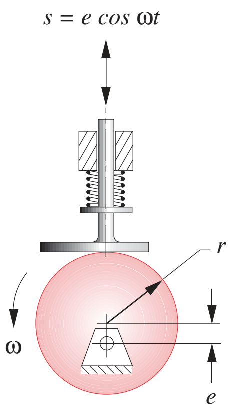

FIGURE P15-2

Problems 15-12 to 15-14, 15-26, 15-28 to 15-32

†‡15-11 Repeat Problem 15-7 for a double-dwell cam to move the roller follower from 0 to 2 inches in 45° with cycloidal motion, dwell for 150°, fall 2 inches in 90° with modified sine motion, and dwell for the remainder. Cam speed is 200 rpm. Choose a suitable spring rate and preload to maintain follower contact. Select a spring from Appendix D. Assume a damping ratio of 0.15.

†‡15-12 The cam in Figure P15-2 is a pure eccentric with eccentricity e = 20 mm and turns at 200 rpm. Follower mass = 1 kg. The spring has a rate of 10 N/m and a preload of 0.2 N. Find the follower force over one revolution. Assume a damping ratio of 0.10. If there is follower jump, redefine the spring rate and preload to eliminate it.

†‡15-13 Repeat Problem 15-12 using a cam with a 20-mm symmetric double harmonic rise and fall (180° rise and 180° fall). See Chapter 8 for cam formulas.

†‡15-14 Repeat Problem 15-12 using a cam with a 20-mm 3-4-5-6 polynomial rise and fall (180° rise and 180° fall). See Chapter 8 for cam formulas.

†‡15-15 Design a double-dwell cam to move a 50-mm-dia roller follower of mass = 2 kg from 0 to 45 mm in 60° with modified sine acceleration, dwell for 120°, fall 45 mm in 90° with 3-4-5 polynomial motion, and dwell for the remainder. The total cycle must take 1 sec. Size a return spring and specify its preload to maintain contact between the cam and follower. Select a spring from Appendix D. Calculate and plot the dynamic force and torque. Assume damping of 0.25 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

†‡15-16 Design a single-dwell cam using polynomials to move a 50-mm-dia roller follower of mass = 10 kg from 0 to 25 mm in 60°, fall 25 mm in 90°, and dwell for the remainder. The total cycle must take 2 sec. Size a return spring and specify its preload to maintain contact between the cam and follower. Select a spring from Appendix D. Calculate and plot the dynamic force and torque. Assume damping of 0.15 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

†‡15-17 Design a four-dwell cam to move a 50-mm-dia roller follower of mass = 3 kg from 0 to 40 mm in 40°, dwell for 100°

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

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‡ These problems are suited to solution using program DYNACAM.

dwell for 40°, fall 10 mm in 30°, and dwell for the remainder. The total cycle must take 10 sec. Choose suitable programs for the rise and fall to minimize dynamic forces and torques. Size a return spring and specify its preload to maintain contact between the cam and follower. Calculate and plot the dynamic force and torque. Assume damping of 0.25 times critical. Repeat for a form-closed cam. Compare the dynamic force, torque, and natural frequency for the form-closed design and the force-closed design.

†‡15-18 Design a cam to drive an automotive valve train whose effective mass is 0.2 kg. ζ = 0.3. Valve stroke is 12 mm. Roller follower is 10 mm diameter. The open-close event occupies 160° of camshaft revolution; dwell for remainder. Use one or two polynomials for the rise-fall event. Select a spring constant and preload to avoid jump to 3500 rpm. Fast opening and closing and maximum open time are desired.

†‡15-19 Figure P15-3 shows a cam-follower system that drives slider 6 through an external output arm 3. Arms 2 and 3 are both rigidly attached to the 0.75-in-dia shaft X-X, which rotates in bearings that are supported by the housing. The pin-to-pin dimensions of the links are shown. The cross sections of arms 2, 3, and 5 are solid, rectangular 1.5 x 0.75 in steel. The ends of these links have a full radius equal to one-half of the link width. Link 4 is 1-in-dia x 0.125 wall round steel tubing. Link 6 is a 2-in-dia x 6-in-long solid steel cylinder. Find the effective mass and effective spring constant of the follower train referenced to the cam-follower roller if the spring at A has a rate of 150 lb/in with a preload of 60 lb. Then determine and plot the kinetostatic follower force and camshaft torque over one cycle if the cam provides a 3-4-5 polynomial double-dwell angular motion to roller arm 2 with a rise of 10° in 90 camshaft degrees, dwell for 90°, fall 10° in 90°, and dwell for the remainder. The camshaft turns 100 rpm.

†‡15-20 Repeat problem 15-19 for a cam that provides a double-dwell cycloidal displacement rather than a 3-4-5 polynomial displacement.

†‡15-21 A single-dwell cam-follower system similar to that shown in Figure 15-1a provides a two-segment polynomial for a rise of 35 mm in 75°, a fall of 35 mm in 120°, and a dwell for the remainder of the cycle. Using equations 15.8 and 15.10, calculate and plot the dynamic force and torque for one cycle if the roller follower train weighs 2.34 N,

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM.

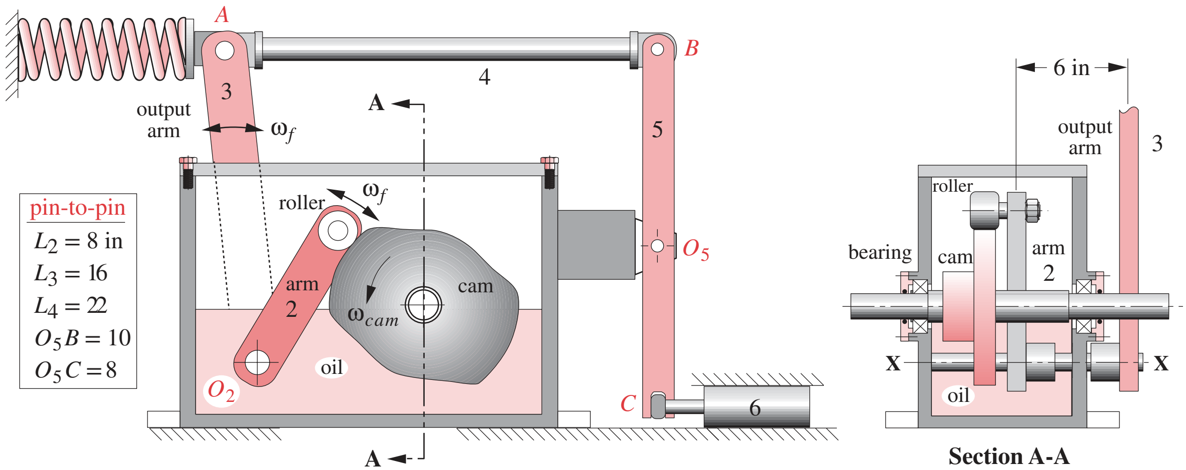

FIGURE P15-3
Problem 15-19 to 15-20

15.11 VIRTUAL LABORATORY

View the downloadable video *Cam_machine_virtual_laboratory.mp4*

View the video (21:28)

Open the file *Cam_Virtual_Lab.zip* and follow the instructions as directed by your professor. Focus on the dynamic force measurements.

View the lab handout

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM.

the system has a damping ratio of $\zeta = 0.06$, and the spring has a rate of 1.5 N/mm with a preload of 10 N. The cam turns at 20 rpm.

†‡15-22 A single-dwell cam-follower system similar to that shown in Figure 15-1a provides a two-segment polynomial for a rise of 35 mm in 75°, a fall of 35 mm in 120°, and a dwell for the remainder of the cycle. Using equations 15.8, size a return spring and specify its preload to maintain contact between the cam and follower. Then calculate and plot the dynamic force for one cycle if the roller follower train weighs 3.55 N, the system has a damping ratio of $\zeta = 0.06$, and the cam turns at 100 rpm.

†‡15-23 A single-dwell cam-follower system similar to that shown in Figure 15-1a provides a constant velocity to the follower of 100 mm/sec for 2 sec then returns to its starting position with a total cycle time of 3 sec. Using equations 15.8 and 15.10, calculate and plot the dynamic force and torque for one cycle if the roller follower train weighs 4.5 N, the system has a damping ratio of $\zeta = 0.06$, and the spring has a rate of 2.5 N/mm with a preload of 50 N.

†‡15-24 A single-dwell cam-follower system similar to that shown in Figure 15-1a provides a constant velocity to the follower of 100 mm/sec for 2 sec then returns to its starting position with a total cycle time of 3 sec. Using equations 15.8, size a return spring and specify its preload to maintain contact between the cam and follower. Then calculate and plot the dynamic force for one cycle of the roller.

†15-25 As stated in Section 15.2, "A common rule of thumb is to design the (cam-follower) system to have a fundamental frequency ω_n at least ten times the highest forcing frequency expected in service..." Since this is not always possible, calculate and plot the amplitude ratio resulting from $5 \le \omega\omega_n/\omega\omega_f \le 10$ for damping ratios of 0, 0.02, 0.04, 0.06, 0.08, and 0.10.

†15-26 A cam-follower system similar to that shown in Figure P15-2 has a lumped mass of 0.1 kg, lumped stiffness of 0.10 N/mm, and a damping coefficient of 0.40 kg/s. If the forcing frequency is 50 rpm, determine the resulting amplitude ratio.

†15-27 A cam-follower system similar to that shown in Figure P15-1 has a lumped mass of 2.5 kg, lumped stiffness of 4 N/mm, and a damping coefficient of 12.0 kg/s. If the forcing frequency is 80 rpm, determine the resulting amplitude ratio.

†15-28 Calculate and plot the amplitude ratio of the cam-follower system of Problem 15-26 for a forcing frequency ranging from 0 to 600 rpm.

†15-29 Calculate and plot the phase angle between the applied force and the displacement of the cam-follower system of Problem 15-26 for a forcing frequency ranging from 0 to 600 rpm.

†15-30 The cam in Figure P15-2 is a pure eccentric with eccentricity $e = 1$ in and turns at 200 rpm. The mass of the follower is 0.01 blobs and the damping ratio is 0.10. Select a spring from Appendix D and a preload such that there is no follower jump. The spring must fit in a 5/8-in hole. Hint: start the iteration with a spring rate of 1 lb/in and a preload of 1 lb.

†15-31 Calculate and plot the camshaft torque on the cam-follower system of Problem 15-30 for one revolution of the cam. Use a spring with stiffness of 2 lb/in and a preload of 2 lb.

†15-32 Calculate and plot the camshaft torque on the cam-follower system of Problem 15-26 for one revolution of the cam, which has an offset of $e = 25$ mm. Use a spring with a stiffness of 0.35 N/mm and a preload of 10 N.